

## Part A · Multiple Choice

2 points each

1. **d) the lithosphere**

*p. 191 — the lithosphere includes the crust plus the rigid uppermost mantle. Nearly all plates carry a combination of oceanic and continental lithosphere.*

2. **a) a few centimeters per year**

*p. 191 — the text compares it to the rate at which fingernails grow. The unit summary gives a range of 1 to 10 cm per year.*

3. **b) a mid-ocean ridge with a rift valley between the plates**

*p. 193 — lava cools quickly at the surface to basalt and more slowly at depth to gabbro, so the whole ridge system is igneous. Shallow earthquakes are common there.*

4. **a) it is denser**

*p. 194 — the denser oceanic plate plunges beneath the continental plate along an oceanic trench, producing a subduction zone with intense earthquakes and a continental volcanic arc above.*

5. **b) continent-to-continent convergence**

*p. 196 — neither plate subducts, so the two buckle upward, producing intense folding and faulting rather than volcanic activity. The Alps and Appalachians formed the same way.*

6. **c) transform boundary**

*p. 196 — the Pacific and North American plates grind past each other at about 3 inches a year. Turkey's North Anatolian Fault is another example; its 1999 Izmit rupture killed 17,000 people in 48 seconds.*

## Part B · True or False

1 point each

7. **TRUE**

*p. 197 — the plate moves while the mantle plume stays put, so volcanoes get older with distance from the hotspot. Geologists use these chains to find the direction and speed of plate motion.*

8. **FALSE**

*Most geologic activity — volcanoes, earthquakes, and mountain building — occurs at plate boundaries. Intraplate activity, such as the Yellowstone hotspot, is the small exception. (pp. 192 and 197)*

## Part C · Short Answer

3 points each

9. **Seismograph networks set up in the 1950s and early 1960s to detect atomic bomb tests also recorded every earthquake worldwide. Plotting the epicenters revealed narrow bands that trace the plate boundaries.**

**Rubric:** 1 pt — the seismograph networks were built during the Cold War to detect enemy atomic tests, and incidentally recorded all earthquakes. 1 pt — seismic records locate an earthquake's epicenter, the point on the surface directly above where the quake occurs. 1 pt — plotting those epicenters produces lines that outline the plates; connecting the dots draws the boundaries, which coincide with mid-ocean ridges, trenches, and large faults. (p. 191)

10. A rift valley forms first (the Great Rift Valley in Africa), then fills with water to make linear lakes (the East African lakes), then widens into a sea (the Red Sea), and finally becomes an ocean (the Atlantic).

**Rubric:** 0.75 pt per stage with its example. (1) **Rift valley** — the Great Rift Valley in Africa. (2) **Linear lakes** — the valley fills with water; the elongated lakes of eastern Africa. (3) **Sea** — continued divergence opens a narrow sea; the Red Sea. (4) **Ocean** — the Atlantic Ocean, which the text calls the ultimate divergent boundary, opened when Pangaea broke apart. Full credit for the correct order even if one example is missing. (p. 193)